Arastun Mammadli

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Homepage | **LinkedIn** | **GitHub** | **Scholar**

Profile

I am a first-year Ph.D. student at the University of Exeter expected to graduate in March 2030. I work on agentic AI, where my goal is to improve the coordination, cost-efficiency, and transparency in LLM multi-agent systems. I am also a visiting researcher at University College London NLP working on retrieval and composition of AI agents. Previously, I interned at Microsoft for my master's dissertation, where I worked on Azure AI and LLM-based services.

Education

University of Exeter, United Kingdom
Ph.D. student in Artificial Intelligence under Prof. Shiqiang Wang. May 2026 – Present

Imperial College London, United Kingdom
MSc Applied Computational Science and Engineering, Distinction Oct. 2024 – Oct. 2025

University of Exeter, United Kingdom
BSc Computer Science, Dean's List (top 5%) Sept. 2020 – May 2024

Experience

University of Exeter, Doctoral Student May 2026 – Present

- Fully funded 3.5-year studentship (fees, stipend and research grant).
- Working on selecting and coordinating LLM agents under game-theoretic cooperative and competitive dynamics.

University College London, NLP Group, Visiting Researcher May 2026 – Present

- Working towards a novel search algorithm for agent reranking (AgentProbe) under Prof. Emine Yilmaz.
- Other contributions include a survey (Agent and Tool Search), and a workshop challenge (AgentSearch SIGIR).

University College London, NLP Group, Research Assistant Nov. 2025 – May 2026

- Constructed AgentSearchBench with over 10K+ profiles and 4K+ tasks to evaluate agent search across the web.
- Benchmarked over 25 models including sparse and dense retrievers, cross-encoders, and generative rerankers.

Microsoft, Research Intern June 2025 – Sept. 2025

- Developed a full-stack prototype with Azure AI Services (SK, Search, Language) for a campus energy system.
- Composite evaluation (200+ task scenarios): LLM sensitivity, query complexity, and architectural ablations.

Imperial College London, Internal Projects Nov. 2024 – May 2025

- 4 projects: numerical solvers, deep time-series forecasting, image processing, evolutionary optimisation.
- Collaborated with a team of 8, managed inter-team coordination (incl. task allocation, project sustainability).

Selected Publications

Choosing LLMs for Strategic Agent Interactions
Arastun Mammadli, Herbert Woisetschläger, Jawad Fayaz, Shiqiang Wang
NeurIPS workshop on Foundations of Agentic Systems Theory (FAST), 2026

AgentSearchBench: A Benchmark for AI Agent Search in the Wild
Bin Wu*, Arastun Mammadli*, Xiaoyu Zhang, Emine Yilmaz
Submitted to AAAI'27, arXiv:2604.22436, 2026 *Co-led

Cost-Optimal LLM Routing with Limited User Feedback under User Satisfaction Guarantees
Herbert Woisetschläger, Arastun Mammadli, Ryan Zhang, Shiqiang Wang
Submitted to ICLR'27, arXiv:2606.19376, 2026

Skills

ML Frameworks: Python, PyTorch, vLLM, HF Transformers and PEFT, Sentence Transformers, AutoGen, FAISS.
HPC & MLOps: Slurm, Docker, Podman-HPC, Weights & Biases, CUDA, REST APIs.
Data & Experimentation: Jupyter, NumPy, Pandas, Scikit-learn, SciPy.